

Polynomial Functions

Knowledge

1. For the function, $f(x) = x^4 - 3x^3 - 39x^2 + 83x - 42$,

a) Graph the function. Clearly label the x- and y- intercepts. Show all work!

$$\begin{array}{r|rrrrr} 7 & 1 & -3 & -39 & 83 & -42 \\ \downarrow & & 7 & 28 & -77 & 42 \\ \hline -6 & 1 & 4 & -11 & 6 & 0 \\ \downarrow & & -6 & 12 & -6 & \\ \hline & 1 & -2 & 1 & 0 & \end{array}$$

$$y = (x-7)(x+6)(x^2-2x+1)$$

$$y = (x-7)(x+6)(x-1)^2$$

$$y\text{-int}$$

$$x=0$$

$$f(0) = -42$$

2. If -2 is one root of the equation $kx^4 + (2k+5)x^3 + (3-k) = 0$, then the value of k is:

$$k(-2)^4 + (2k+5)(-2)^3 + 3-k = 0$$

$$16k + (2k+5)(-8) + 3-k = 0$$

$$16k - 16k - 40 + 3 - k = 0$$

$$k = -37$$

3. The zeros of a family of functions are -1 order 2; $\frac{3}{4}$ and 4. Determine an equation, in factored form, for the member of the family that passes through the point (2,18).

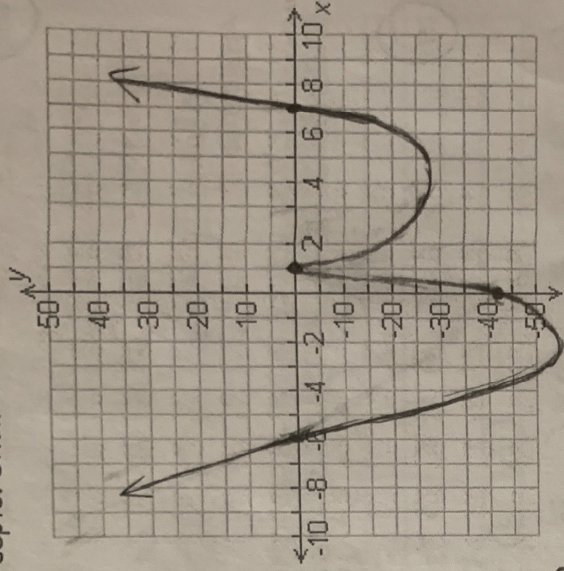
$$y = a(x+1)^2(4x+3)(x-4)$$

$$18 = a(2+1)^2(4(2)+3)(2-4)$$

$$18 = a(9)(11)(-2)$$

$$a = -\frac{1}{11}$$

$$y = -\frac{1}{11}(x+1)^2(4x+3)(x-4)$$



Communication

- Without a calculator, explain why $-3x^{12} - 2012x^6 + 3x^2 + 10 < 15 + 6x^2$ has the solution of $(-\infty, \infty)$. and they are added

$$3x^{12} + 2012x^6 + 3x^2 + 5 > 0$$

It is so because all of the terms are always positive. All of the powers are even. Note that a negative raised to an even power is always positive. Also all of the coefficients are positive, and positive + positive is always positive. $(-a)^{\text{even}} = (-1)^{\text{even}} a^{\text{even}} = a^{\text{even}} > 0$

Application

- What is the equation, in expanded form, for a function with zeros $3 \pm \sqrt{5}i$.

$$f(x) = a(x - (3 + \sqrt{5}i))(x - (3 - \sqrt{5}i))$$

$$f(x) = a(x - 3 - \sqrt{5}i)(x - 3 + \sqrt{5}i)$$

$$f(x) = a((x - 3)^2 - (\sqrt{5}i)^2)$$

$$f(x) = a(x^2 - 6x + 9 + 5)$$

$$f(x) = a(x^2 - 6x + 14)$$

choose any $a \in (-\infty, \infty) \cup (0, \infty)$

eg $a = 1$

$$f(x) = x^2 - 6x + 14$$

- The new ride at Azar's Wonderland is a rollercoaster that starts above the ground but then dives into the water for a period of time. The height of the roller coaster can be modelled by the polynomial function $h(x) = x^4 - 25x^3 + 203x^2 - 599x + 420$. If the x-axis represents the water level, for how long was the ride under water?

$$x^4 - 25x^3 + 203x^2 - 599x + 420 < 0$$

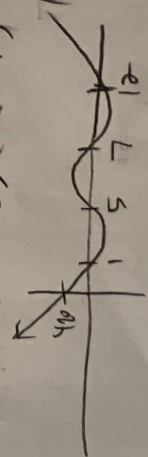
1	1	-25	203	-599	420
	↓	1	-24	179	-420
5	1	-24	179	-420	0
	↓	5	-95	420	
1	-19	84	0		

∴ the ride was under water for 9 seconds

$$(x - 1)(x - 5)(x^2 - 19x + 84) < 0$$

$$(x - 1)(x - 5)(x - 12)(x - 7) < 0$$

$$\text{ANS} = (5 - 1)(12 - 7) = 4 + 5 = 9$$



3. If $f(x) = x^4 + 7x^3 - 4x^2 + 20$ and $g(x) = x^4 + 6x^3 - 11x^2 - 7x + 35$, solve the inequality $f(x) \geq g(x)$ using a chart/table.

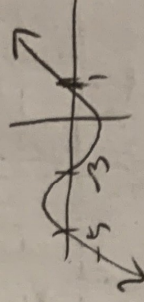
$$x^4 + 7x^3 - 4x^2 + 20 \geq x^4 + 6x^3 - 11x^2 - 7x + 35$$

$$x^3 + 7x^2 + 7x - 15 \geq 0$$

$$(x-1)(x^2 + 8x + 15) \geq 0$$

$$(x-1)(x+3)(x+5) \geq 0$$

$$\begin{array}{r} 1 \quad 1 \quad 7 \quad 7 \quad -15 \\ \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\ 1 \quad 8 \quad 15 \quad 0 \end{array}$$



Intervals	$(-\infty, -5)$	$(-5, -3)$	$(-3, 1)$	$(1, \infty)$
Test Value	-6	-4	0	5
$(x+5)$	-	+	+	+
$(x+3)$	-	-	+	+
$(x+1)$	-	-	-	+
$(x+5)(x+3)(x+1)$	-	+	-	+

Concluding Statement $\therefore f(x) \geq g(x)$ when $x \in [-5, -3] \cup [1, \infty)$

Thinking

1. Determine the value of b , c , and d such that the polynomial $f(x) = x^3 + bx^2 + cx + d$ satisfies all the following:

- when divided by $x+1$ the remainder is -5 .
- when divided by $x-3$ the remainder is 31 .
- when it is divided by $x-2$ and $x+2$ the remainder are the same.

$$f(-1) = -5 = (-1)^3 + b(-1)^2 + c(-1) + d$$

$$-5 = -1 + b - c + d$$

$$-4 = b - c + d \quad (1)$$

$$f(3) = 31 = (3)^3 + b(3)^2 + c(3) + d$$

$$31 = 27 + 9b + 3c + d$$

$$4 = 9b + 3c + d \quad (2)$$

$$f(2) = f(-2)$$

$$(2)^3 + b(2)^2 + c(2) + d = (-2)^3 + b(-2)^2 + c(-2) + d$$

$$8 + 4b + 2c + d = -8 + 4b - 2c + d$$

$$4c = -16 \rightarrow c = -4 \quad (3)$$

when it is divided by $x-2$ and $x+2$ the remainder are the same.

$$\begin{array}{l} b = 3 \\ c = -4 \\ d = -11 \end{array}$$

sub (3) into (1)

$$-4 = b - (-4) + d$$

$$-4 = b + 4 + d$$

$$b + d = -8 \quad (4)$$

sub (3) into (2)

$$4 = 9b + 3(-4) + d$$

$$16 = 9b + d \quad (5)$$

$$(5) - (4)$$

$$8b = 24$$

$$b = 3$$

$$3 + d = -8$$

2. Mr. Azar needs help to solve this polynomial inequality $f(x)$: $343x^3 - 392x^2 + 42x^2 + 8x - 1 > 0$. He used the graphing calculator, graphed it and determined the zeros to be approximately -0.14, 0.14, and exactly 1. Can you help him with the solution? Justify your reasoning.

Degree 4 but only 3 roots $\rightarrow$ find last root by dividing roots that we already know

$$\begin{array}{r|rrrr} 1 & 343 & -392 & 42 & 8 & -1 \\ & \downarrow & & & & \\ & 343 & -49 & -49 & -7 & -1 \\ \hline & 343 & -49 & -7 & 1 & 0 \end{array}$$

$$\begin{aligned} & (x-0.14)(x+0.14) \\ &= x^2 - 0.14^2 \\ &= x^2 - 0.0196 \end{aligned}$$

$$\begin{array}{r} 343x - 49 \\ 343x^3 - 49x^2 - 7x + 1 \\ \hline -1 \end{array}$$

$$\begin{array}{r} -49x^2 - 0.2972x + 1 \\ -1 \end{array}$$

last root

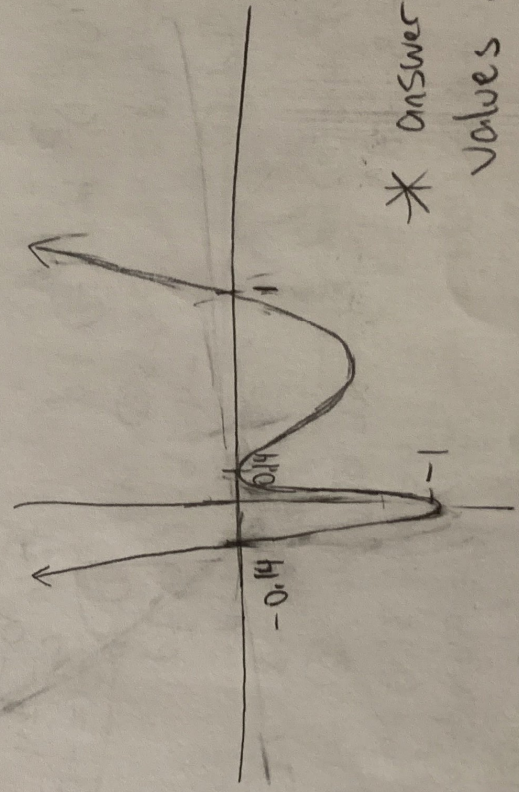
$$\begin{array}{r} -1 - 49x^2 + 0x + 0.9604 \\ \hline \end{array}$$

$$\begin{aligned} 343x - 49 &= 0 \\ x &= \frac{49}{343} \approx 0.14 \end{aligned}$$

$$-0.2772x + 0.0396 \approx 0$$

from inside

$$f(x) \approx (x-1)(x-0.14)^2(x+0.14) = 0$$



$\therefore f(x) > 0$ when

$$x \in (-\infty, -0.14) \cup (1, \infty) *$$

* answer not exact because exact values of roots are not given.

$\rightarrow$ Need to use graphing calculator